

Walmart Demand Forecasting & Time Series Modeling

Interview-Ready Study Guide

1. Project Overview & Business Framing

Executive Summary

This project addresses a classic retail demand forecasting problem: predicting weekly Walmart sales across 45 stores from 2010 to 2012 using historical sales, calendar effects, and macroeconomic variables.

Demand forecasting is critical because it improves inventory holding cost, safety stock optimization, working capital efficiency, stockout prevention, and labor scheduling.

2. Tech Stack Selection & Justification

Tool	Why It Was Used	Technical Justification
Python	Core forecasting workflow	Clean scripting, rapid experimentation, reproducible pipelines
VS Code + antipater/Devify	Development environment	Isolated package versions and reproducible setup
Pandas & NumPy	Temporal indexing and feature engineering	Large engineering, rolling windows, and fast vectorized operations
Matplotlib & Seaborn	Visualization	Trend analysis, holiday lifts, residual diagnostics
Statsmodels SARIMAX	Classical statistical baseline	State-space structure with exogenous effects
Meta Prophet	Production benchmark	Additive decomposition with seasonal and holiday components
PyTorch LSTM	Deep sequence model	Captures nonlinear temporal patterns
Scikit-Learn	Evaluation and scaling	MinMaxScaler, RMSE, MAE, MAPE, WAPE

3. Exploratory Data Analysis (EDA) & Statistical Findings

Mean Weekly Sales: \$1.05M

Median Weekly Sales: \$960K

Distribution: right-skewed with pronounced holiday-driven peaks

Macro Trends & Seasonality: Q4 demand spikes from Thanksgiving, Black Friday, and Christmas drove weekly sales above \$80M.

Holiday Lift: average holiday week sales were approximately \$1.12M versus \$1.04M for non-holiday weeks, a lift of about 7.8%.

Store variance: Store 20 averaged \$2.11M weekly while Store 33 averaged \$259.8K; this is roughly an 8x spread.

External feature correlations: CPI = -0.07, Fuel Price = +0.01, Unemployment = -0.11, which shows that linear relationships alone are insufficient.

4. Feature Engineering & Chronological Splitting Strategy

Random K-fold cross-validation is inappropriate for time series because future observations leak into training data.

A strict chronological 80/20 split is mandatory to simulate real deployment conditions.

Lag features included Lag_1, Lag_2, Lag_4, and Lag_52 to capture short-term dependence and annual seasonality.

Rolling features such as 4-week moving averages and standard deviations were computed using only historical values to preserve causality.

5. Model Architecture, Benchmark & Deep Statistical Failure Analysis

Model	RMSE	MAE	MAPE
Meta Prophet	\$89,364.40	\$64,160.55	4.07%
PyTorch LSTM	\$106,086.94	\$91,002.57	5.97%
SARIMAX	\$518,490.72	\$480,853.96	31.02%

Why Prophet Won

Prophet achieved the best result because its additive decomposition captures trend, annual seasonality, and holiday effects correctly. With a short weekly series, the model benefits from regularization and a strong structural prior.

Why SARIMAX Failed

SARIMAX was statistically underpowered because it tried to estimate a 52-week seasonal pattern with only about 115 training points, creating parameter explosion and numerical instability.

Why LSTM Placed Second

The LSTM is a strong nonlinear model, but this problem is data-starved; with limited weekly observations, deep sequence models have higher variance than regularized additive models.

6. Model Trade-Off Matrix

Model	Accuracy	Interpretability	Compute Cost	Data Appetite
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Prophet	Best	High	Low to Moderate	Good on short history
LSTM	Good	Low	High	Needs more data
SARIMAX	Poor here	Moderate to High	Low to Moderate	Fragile on short seasonal series

7. Production System Architecture & MLOps Design

- Retraining cadence: monthly batch retraining scheduled automatically.
- Drift monitoring: trigger if rolling 2-week MAPE exceeds 10%.
- Inference pipeline: batch-scoring weekly store-level demand and exporting replenishment orders.
- Operational review: compare forecast vs actuals after every weekly cycle.

8. Top 10 Technical Interview Questions

- Why did you choose a chronological split over cross-validation?
- Why did SARIMAX struggle compared to Prophet on this dataset?
- How did you prevent data leakage when creating rolling window features?
- What are the trade-offs between LSTM and Prophet for enterprise forecasting?
- How would you translate 4.07% MAPE into business value?
- How would the architecture scale with 100,000 SKUs?
- What metrics would you monitor for concept drift?
- Why are CPI and Fuel Price poorly correlated with sales?
- How did you handle the variance between Store 20 and Store 33?
- If given another month, what improvements would you implement?

Prepared for technical interview readiness and executive-level communication.